

● TECHNICAL SECTION

Equivariant Neural Networks

Experiments on MNIST & the 6/9 Problem

Part 1

4 models on rotated/unrotated MNIST

Part 2

The 6/9 problem & group restriction

Experimental Setup: 4 Models

Train 4 models, and evaluate on unrotated MNIST / rotated MNIST.

CNN

- trained on unrotated MNIST
- trained on rotated MNIST

C₄ ENN

- trained on unrotated MNIST
- trained on rotated MNIST

Each trained model is evaluated on BOTH unrotated MNIST and rotated MNIST.



NOTE TO REPORTER

These parts are necessary background — in the actual report some parts can be omitted accordingly.

Definitions & Evaluation

C_4 ENN

C_4 : the C_4 group. ENN: Equivariance Neural Network.

MNIST

A dataset of handwritten-digit pictures, each labeled with the digit it shows. Both train and test data included.

rotated

Before train/test, the picture was rotated in advance.

Evaluation

Evaluate the trained model: test the accuracy of correctly distinguishing the digit given a test picture.

Footnote: why C_4 ?

For the grid $\mathbb{Z}^2$, only subgroups of D_4 are exact symmetries. (ref: chap. 2.8)



NOTE TO REPORTER

Background slides — can be skipped or compressed when integrating into the final deck.

△ MAIN RESULT

Comparison ① — CNN vs C_4 ENN, both trained on rotated MNIST

Tested on rotated MNIST

CNN (train rot)

96.65%

accuracy on rotated MNIST

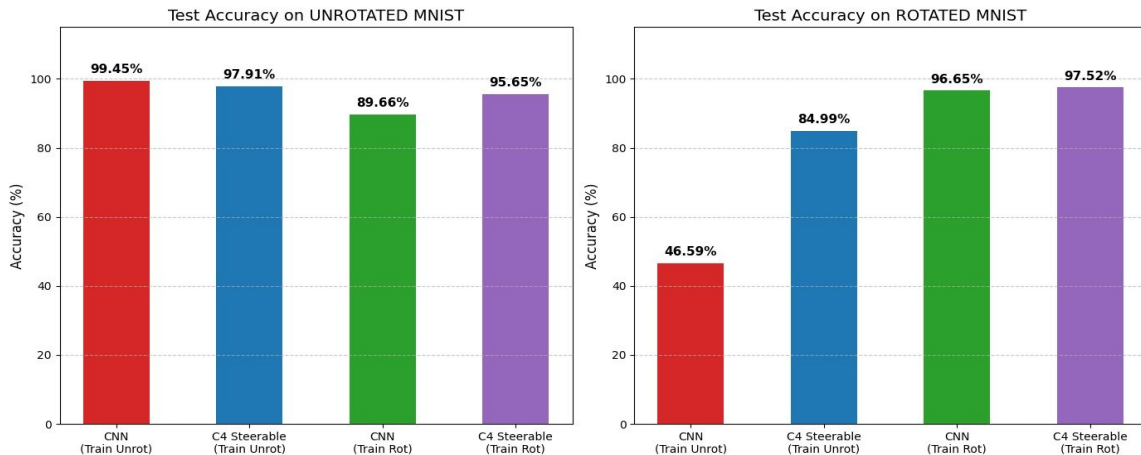
C_4 ENN (train rot)

97.52%

accuracy on rotated MNIST

Takeaway C_4 steerable has the better accuracy — meaning the inherited rotational symmetry built into the network does make models perform better at distinguishing rotated data.

Other Comparisons — ② ③ ④



② Train unrot → Test rotated

CNN 46.59% vs C₄ ENN 84.99%

CNN learns almost nothing about the rotational feature of digits. C₄ ENN still performs well — it doesn't learn rotation from data, but inherits the rotational symmetry by construction.

③ Train unrot → Test unrotated

CNN 99.45% vs C₄ ENN 97.91%

The compulsory group restriction on C₄ ENN reduces learnable parameters — making it slightly less expressive on the standard (unrotated) task.

④ Train rot → Test unrotated

CNN 89.66% vs C₄ ENN 95.65%

Benefiting from built-in symmetry structure, ENN can spend more resource learning the digit itself rather than rotated features. CNN has to separate some resource to learn rotated features by rote. → Capacity Dilution

Capacity Dilution

When CNN is forced to learn rotational invariance from data, some of its model capacity is spent on memorizing rotation by rote — leaving less capacity for the actual digit-classification task. ENN avoids this by encoding rotational symmetry directly in its structure.



NOTE TO REPORTER

Too long for 5-min talk — skip or skim.

The 6/9 Problem

Original "6"

6

Rotated "6" → looks like "9"



180°

rotate

9

Why ENN fails here

ENN with rotational symmetry can only learn *in the sense of rotation*.

In the sense of rotation, 6 and 9 look alike.

Therefore, it is easy for an ENN to mistake 6 for 9.

Footnote

The problem generalizes to other digits (e.g. 5 and 7), to tasks beyond digit classification, and to broader symmetry groups beyond rotation.

Solution: Group Restriction

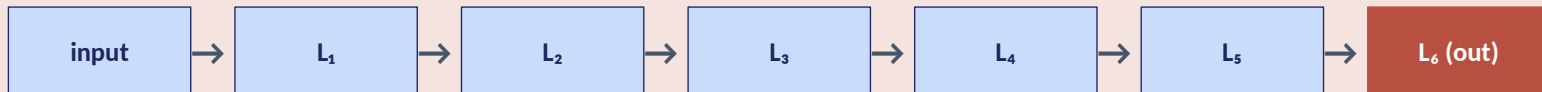
Idea Keep the C_4 structure in the first 5 layers (same as traditional C_4 ENN — invariant under 90° , 180° , 270° , 360° rotation), then restrict the group to the trivial group $\{e\}$ in the 6th layer (no symmetry restriction).

① Traditional C_4 ENN



$\text{input} \Leftrightarrow \text{output} : C_4 \text{ operation invariant}$

② Refined $C_4 | \{e\}$ ENN



$\text{input} \Leftrightarrow L_5 : C_4 \text{ operation invariant}$ | $L_5 \Leftrightarrow L_6 : \{e\} \text{ operation invariant (group restriction } C_4 \rightarrow \{e\})$

Footnote ① In practice we also need to claim the representation of the group, and there is a fully-connected classifier at the end — omitted here for simplicity. **Footnote ②** For details on group restriction, see ref chap. 2.7.

Intuitive Perspective

Different layers, different scales

1

Deeper layer = broader receptive field

The deeper a layer is, the broader its receptive field.

2

Different depth $\leftrightarrow$ different scale

Requiring different symmetries between layers of different depth is essentially requiring different symmetries at different scales.

3

Sacrifice large scale, keep small scale

In $C_4 \mid \{e\}$, we sacrifice the large-scale symmetry while maintaining the small-scale C_4 symmetry.

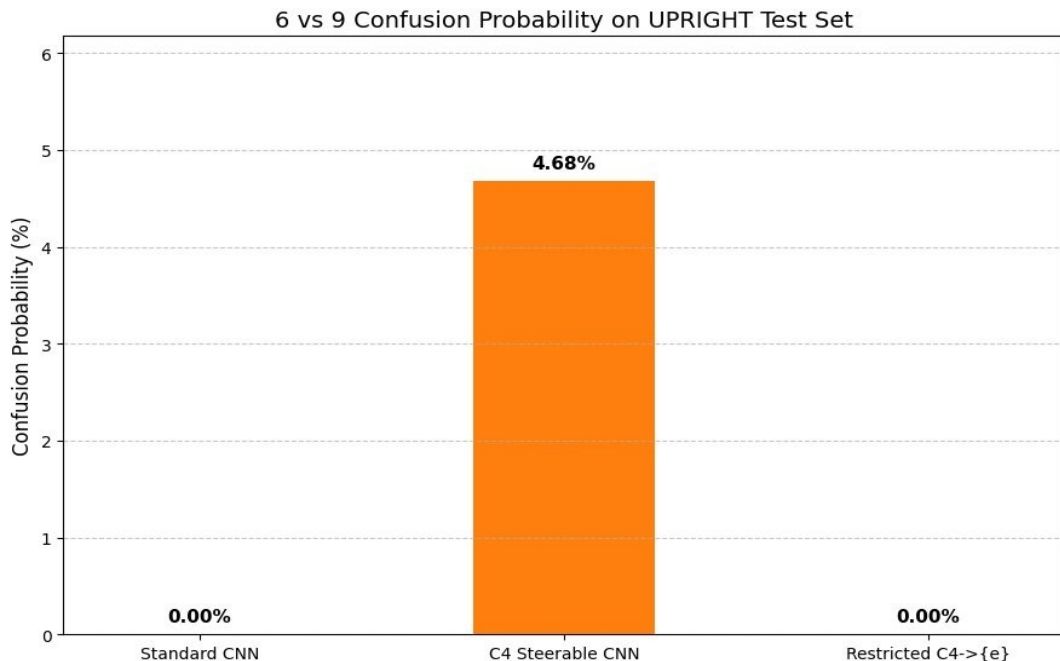
KEY POINT Group restriction is a technique to mitigate the 6/9 problem.



NOTE TO REPORTER

For timing: reporter can skim the technical parts — focus on the intro and the result. (Ref chap. 2.7 for full details.)

Result: 6/9 Confusion Probability



What this shows

Traditional C_4 ENN

4.68%

Suffers from the 6/9 problem.

Refined $C_4 | \{e\}$

0.00%

Group restriction eliminates it.

Standard CNN

0.00%

Doesn't have rotational symmetry, so no confusion.

Footnote The 6/9 problem in traditional C_4 ENN is not too severe in practice — handwritten 6 and 9 still have differences beyond pure rotation. But $C_4 | \{e\}$ and CNN don't have the problem at all.

References

All chapter references in this section point to:

General E(2)-Equivariant Steerable CNNs

Maurice Weiler, Gabriele Cesa · NeurIPS 2019

<https://arxiv.org/abs/1911.08251>

Specific chapters cited in this section:

- **Chapter 2.8** — rationale for choosing the C_4 group on the $\mathbb{Z}^2$ grid
- **Chapter 2** — inherent rotational features of C_4 ENN constructions
- **Chapter 2.7** — full details of group restriction